Xiaoyang Lu

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Research Area & Focus

Computer Architecture & Systems, with a focus on **data-centric system design** for memory-intensive AI/HPC workloads.

- **Concurrency-Aware Memory-System Optimization:** Quantifying and exploiting memory-access concurrency to guide cache management, prefetching, and memory-hierarchy optimization [C1, C8, C10, C11, C12, C14, C16].
- **Performance & I/O Modeling:** Modeling memory and data-movement costs to guide architecture and system decisions across AI/HPC workloads [C4, C5, C13, J3, P2].
- **Learning-Assisted System Optimization:** Using lightweight machine-learning methods, including reinforcement learning, to predict memory behavior and guide system optimization [C2, C7, C8, C11].
- **Data-Centric AI/HPC Architectures:** Reducing data movement through PIM/near-data execution and memory-centric system design for emerging AI workloads [C3, C4, C9, C15, J1, J2, P1].

Academic Appointments

Research Assistant Professor June 2024 – Present
Illinois Institute of Technology, Gnosis Research Center (GRC) Chicago, IL

- Initiate and lead research projects in data-centric computer systems, spanning concurrency-aware memory-system optimization, performance modeling, and memory-centric architectures for AI and HPC workloads.
- Develop model-guided performance-matching methods that translate workload data-access and data-movement requirements into architecture, data-placement, resource-selection, and heterogeneous-execution decisions.
- Co-advise and mentor Ph.D. students across memory-centric and AI systems research, guiding projects from problem formulation and system design through evaluation and publication, including multi-institutional collaborations.

Education

Illinois Institute of Technology Chicago, IL
Ph.D. in Computer Science Aug 2017 – May 2024
Advisor: *Professor Xian-He Sun*
Thesis: *Utilizing Concurrent Data Accesses for Data-Driven and AI Applications*

New York University New York, NY
M.S. in Computer Engineering Aug 2015 – May 2017

Zhejiang University Hangzhou, China
B.E. in Electronic Science and Technology Aug 2011 – Jul 2015

Prior Research Experience

Research Assistant Jan 2020 – May 2024
Illinois Institute of Technology Chicago, IL

- Developed a computer-architecture research direction in concurrency-aware memory systems, spanning cache management, prefetching, and memory-performance modeling for data-intensive workloads.
- Developed analytical and learning-assisted methods that quantify memory-access concurrency and translate it into optimization decisions across the memory hierarchy.

Research Aide May 2020 – Aug 2020
Argonne National Laboratory Lemont, IL

- Characterized disaggregated-memory performance, focusing on memory-access bottlenecks, interference, and performance variability.

Sponsored Research & Collaborations

- **NSF SHF: Small - Optimization of Memory Architectures: A Foundation Approach (CCF-2008907).**
  Role and contributions: Technical lead in developing concurrency-aware memory-performance modeling and optimization, including prefetching, cache partitioning, replacement, and holistic cache control.
  Representative publications: [C11, C12, C13, C14, C16, J3].

- **NSF CSR: Medium - Towards a Unified Memory-Centric Computing System with Cross-Layer Support (CNS-2310422).**
  Role and contributions: Technical lead in developing memory-centric systems and execution methods for data-intensive AI workloads, including sparse-matrix acceleration, I/O analysis for long-sequence attention, and PIM-based execution.
  Representative publications: [C2, C4, C5, C10].

- **DOE Genesis Mission - PORTUS (Phase I; PNNL-led): Automatic Performance Portability for Distributed Scientific Workflows on the Genesis Mission Platform.**
  Role and contributions: Co-lead GRC's workflow performance-modeling effort, developing dataflow-aware models and resource-matching methods to guide storage selection, data placement, and task parallelism across heterogeneous HPC and cloud resources, in collaboration with PNNL.

Submitted Research Proposals (Under Review)

- **NSF CSR - Semantics-Driven Proactive Memory-Centric Computing for Stateful AI.**
  Co-PI; submitted 2026.

- **Amazon Research Awards - Pattern-Aware Self-Improving Agents for Dynamic Block-Sparse Attention on AWS Trainium.**
  PI; submitted 2026.

Conference Publications

HPCA'27 [C1] Haozhe Song, **Xiaoyang Lu**, Shuibing He, Xian-He Sun. *Rethinking Tiered Memory Management through Memory Access Concurrency*. In the 33rd IEEE International Symposium on High-Performance Computer Architecture (HPCA), 2027.

PACT'26 [C2] Lihan Hu, **Xiaoyang Lu**, Xian-He Sun, Peng Jiang. *Improving Data Reuse across Blocks for Efficient Block-Sparse Transformers on GPUs*. In the 35th International Conference on Parallel Architectures and Compilation Techniques (PACT), 2026.

CCMCC'26 [C3] Jinlin Wu, Ferit Ozdaban, Selvanathan Sadhasivam, **Xiaoyang Lu**, Yunhui Guo, Minxuan Zhou. *HyPIM: Accelerating Hyperbolic Machine Learning via Processing-In-Memory*. In the 2nd IEEE Cross-Disciplinary Conference on Memory-Centric Computing (CCMCC), 2026.

IPDPS'26 [C4] **Xiaoyang Lu**, Lihan Hu, Hongrui Huang, Peng Jiang, Xian-He Sun. *I/O-Aware PIM Acceleration for Long-Sequence LLM Inference with Hybrid Sparse Attention*. In the 40th IEEE International Parallel & Distributed Processing Symposium (IPDPS), 2026.

ASPLOS'26 [C5] **Xiaoyang Lu**, Boyu Long, Xiaoming Chen, Yinhe Han, Xian-He Sun. *I/O Analysis is All You Need: An I/O Analysis for Long-Sequence Attention*. In the 31st ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), 2026.

DATE'26 [C6] Vadim Biryukov, **Xiaoyang Lu**, Zirui Liu, Kaixiong Zhou, Xian-He Sun. *Zion: A Comprehensive, Adaptive, and Lightweight Hardware Prefetcher*. In the Design, Automation and Test in Europe Conference (DATE), 2026.

MICRO'25 [C7] Haoran Geng, **Xiaoyang Lu**, Yuezhi Che, Ziang Tian, Dazhao Cheng, Xian-He Sun, Michael Niemier, X. Sharon Hu. *COSMOS: RL-Enhanced Locality-Aware Counter Cache Optimization for Secure Memory*. In the 58th IEEE/ACM International Symposium on Microarchitecture (MICRO), 2025.

GLSVLSI'25 [C8] Yuping Wu, **Xiaoyang Lu**, Xiaoming Chen, Yinhe Han, Xian-He Sun. *Concurrency-Aware Cache Miss Cost Prediction with Perceptron Learning*. In the 35th ACM Great Lakes Symposium on VLSI (GLSVLSI), 2025.

ICCD'24 [C9] Liang Yan, **Xiaoyang Lu**, Xiaoming Chen, Sheng Xu, Xingqi Zou, Yinhe Han, Xian-He Sun. *AceMiner: Accelerating Graph Pattern Matching using PIM with Optimized Cache System*. In the 42nd IEEE International Conference on Computer Design (ICCD), 2024.

ASPLOS'24 [C10] **Xiaoyang Lu**, Boyu Long, Xiaoming Chen, Yinhe Han, Xian-He Sun. *ACES: Accelerating Sparse Matrix Multiplication with Adaptive Execution Flow and Concurrency-Aware Cache Optimizations*. In the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), 2024.

HPCA'24 [C11] **Xiaoyang Lu**, Hamed Najafi, Jason Liu, Xian-He Sun. *CHROME: Concurrency-Aware Holistic Cache Management Framework with Online Reinforcement Learning*. In the 30th IEEE International Symposium on High-Performance Computer Architecture (HPCA), 2024.

HPCA'23 [C12] **Xiaoyang Lu**, Rujia Wang, Xian-He Sun. *CARE: A Concurrency-Aware Enhanced Lightweight Cache Management Framework*. In the 29th IEEE International Symposium on High-Performance Computer Architecture (HPCA), 2023.

WSC'22 [C13] Hamed Najafi, **Xiaoyang Lu**, Jason Liu, Xian-He Sun. *A Generalized Model For Modern Hierarchical Memory System*. In the Winter Simulation Conference (WSC), 2022.

ICCD'21 [C14] **Xiaoyang Lu**, Rujia Wang, Xian-He Sun. *Premier: A Concurrency-Aware Pseudo-Partitioning Framework for Shared Last-Level Cache*. In the 39th IEEE International Conference on Computer Design (ICCD), 2021.

ISLPED'21 [C15] Liang Yan, Mingzhe Zhang, Rujia Wang, Xiaoming Chen, Xingqi Zou, **Xiaoyang Lu**, Yinhe Han, Xian-He Sun. *CoPIM: A Concurrency-aware PIM Workload Offloading Architecture for Graph Applications*. In the IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED), 2021.

ICCD'20 [C16] **Xiaoyang Lu**, Rujia Wang, Xian-He Sun. *APAC: An Accurate and Adaptive Prefetch Framework with Concurrent Memory Access Analysis*. In the 38th IEEE International Conference on Computer Design (ICCD), 2020.

Journal Publications

TCAD'25 [J1] Liang Yan, **Xiaoyang Lu**, Sheng Xu, Xiaoming Chen, Xingqi Zou, Yinhe Han, Xian-He Sun. *ProMiner: Enhancing Locality, Parallelism, and Offloading for Graph Mining on Processing-in-Memory Systems*. IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 44(12): 4577–4590, 2025.

CAL'25 [J2] Liang Yan, **Xiaoyang Lu**, Xiaoming Chen, Yinhe Han, Xian-He Sun. *Pyramid: Accelerating LLM Inference with Cross-Level Processing-in-Memory*. IEEE Computer Architecture Letters, 24(1): 121–124, 2025.

JCST'23 [J3] Xian-He Sun, **Xiaoyang Lu**. *The Memory-Bounded Speedup Model and Its Impacts in Computing*. Journal of Computer Science and Technology, 38(1): 64–79, 2023.

Preprints

arXiv'26 [P1] **Xiaoyang Lu**, Belthangady Akash Vi Narayana Pai, Xian-He Sun. *DynaNDE: Dynamic Near-Data Expert Scheduling for Batched MoE Inference*. arXiv preprint, 2026.

arXiv'26 [P2] Haoran Geng, Tomas Sousa Pereira, **Xiaoyang Lu**, Xian-He Sun, Michael Niemier, X. Sharon Hu. *VIPER: Architecture-Aware Performance Modeling for Processing-in-Memory Design-Space Exploration*. arXiv preprint, 2026.

Research Mentoring

Ph.D. Co-Advising

- **Vadim Biryukov**, Ph.D. candidate, Illinois Institute of Technology, 2023–Present; co-advised with Prof. Xian-He Sun; *CPU architecture and memory systems*; **1× DATE** [C6].
- **Anwar Benhnini**, Ph.D. student, Illinois Institute of Technology, 2026–Present; co-advised with Prof. Xian-He Sun; *memory-centric architectures for AI*.
- **Alexander Ichetovkin**, Ph.D. student, Illinois Institute of Technology, 2026–Present; co-advised with Prof. Xian-He Sun; *memory-system performance modeling*.
- **Xulu Chu**, Ph.D. student, Oregon State University, 2026–Present; co-advised with Prof. Wenqian Dong; *wafer-scale architectures for AI*.

Collaborative Ph.D. Mentoring

- **Haoran Geng**, Ph.D. candidate, University of Notre Dame, 2023–Present; mentored in collaboration with Prof. X. Sharon Hu; *memory-system optimization and processing-in-memory*; **1× MICRO** [C7], **1 preprint** [P2].

- **Lihan Hu**, Ph.D. candidate, University of Iowa, 2024–2026; mentored in collaboration with Prof. Peng Jiang; *sparsity-aware architectures for AI*; **1× PACT** [C2], **1× IPDPS** [C4].

M.S. and Undergraduate Mentoring

- **Hongrui Huang**, M.S. student, Columbia University, 2025–2026; *processing-in-memory for AI acceleration*; **1× IPDPS** [C4].
- **Belthangady Akash Vi Narayana Pai**, M.S. student, Illinois Institute of Technology, 2025–2026; *near-memory processing for AI*; **1 preprint** [P1].
- **Max Han**, undergraduate student, University of Illinois Urbana-Champaign, 2024–2026; *cache and memory-system optimization.*

Teaching Experience

Guest Lectures

- Fall 2026, ECE 6210 Machine Intelligence, George Washington University: "Reinforcement Learning for Intelligent Memory Systems".
- Spring 2025, CS 550 Advanced Operating Systems, Illinois Institute of Technology: "Data-Centric Optimizations for LLMs".
- Fall 2024, CS 546 Parallel and Distributed Processing, Illinois Institute of Technology: "Introduction to Parallel Processing".
- Spring 2022, CS 570 Advanced Computer Architecture, Illinois Institute of Technology: "GPU Architectures".

Teaching Assistant — Aug 2017 – May 2022

Illinois Institute of Technology — Chicago, IL

- Supported computer science courses including Introduction to Advanced Studies I (CS 401), Software Engineering (CS 487), Advanced Operating Systems (CS 550), Parallel and Distributed Processing (CS 546), and Advanced Computer Architecture (CS 570).
- Designed laboratory exercises, assignments, examinations, and course materials; led laboratory sessions and technical discussions for classes of up to 60 students; and mentored student course projects.

Teaching Interests: Computer Architecture, AI/ML Systems, Operating Systems, and Parallel and Distributed Systems.

Professional Service

Editorial Service

- Guest Editor, *IEEE Internet of Things Journal*, Special Issue on Integrated Sensing, Memory, Communication, and Computation for Large-Scale AI Based IoT Systems, 2026.

Conference Service

- **Program Committee Member:** DAC'27, EuroSys'27, HPCA'27, DAC'26, IEEE Cloud Summit'26, ICCD'25.
- **External Review Committee Member:** MLSys'26.
- **Shadow Program Committee Member:** EuroSys'26.
- **Session Chair:** PACT'26.

Selected Journal Reviewing

- ACM TACO; IEEE TC, TPDS, TII, TNSE, TCE; JSA; MICPRO; FGCS; Neurocomputing; ESWA; Device; MEMORI; Simulation.

Invited Talks

- *Toward Data-Centric Computing: Data-Movement Analysis, Concurrency-Aware Optimization, and Memory-Centric Architecture.* Invited talk, Workshop on Extreme-Scale Storage and Analysis (ESSA), 2026.

Academic Honors and Awards

- 2024 DAC PhD Forum Travel Award
- 2024 Illinois Institute of Technology Computer Science Department Best Student Paper Award (2023–2024)
- 2024 Illinois Institute of Technology College of Computing Best Poster Award
- 2024 ASPLOS Student Travel Award
- 2023 Top 100 Chips Achievements (2022–2023)
- 2023 HPCA Student Travel Award
- 2015 New York University Scholarship
- 2015 Zhejiang University Excellent Bachelor Thesis Award